Master of Science in Artificial Intelligence, 2026-2028



Knowledge Representation and Reasoning

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Lecture 5

Bayesian Networks Learning

- Maximum Likelihood Estimation
- Bayesian parameter learning

Is BN learning still relevant in 2026?

Yes but not for raw prediction. Deep networks win on images, text and audio. BN learning matters where a model must be **read, questioned and trusted**

Readable models

Clinical decision support, reliability, risk. The EU AI Act asks high-risk systems for transparency and human oversight; a BN is auditable by design.

Causal questions

“What happens if we treat?” needs a graph. Structure learning is the core of causal discovery, an active field today.

Ideas that travel

Likelihood, priors, and EM are the same ideas used to train today’s neural models.

What does it mean to learn a BN?

Nature generates data from an unknown distribution **p***.
We see **N** samples $D = \{x^{(1)}, \ldots, x^{(N)}\}$, assumed **i.i.d.**

Parameters θ – the numbers in every CPT

Structure G – graph structure

Goal: find (G, θ) whose distribution **p** is as close as possible to p*.

"Close" for what?

Density estimation — answer any query later

Prediction — one target, e.g. diagnosis

Knowledge discovery — the graph itself is the result

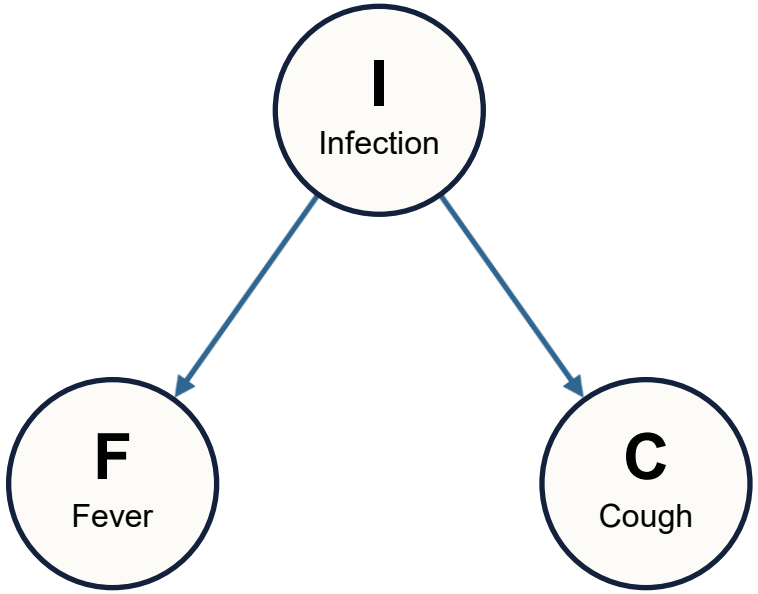
Data instances are *independent and identically distributed* (**iid**) – they are **generated independently** and from **identical unknown distributions**

What do we learn?

Four settings, from easy to hard

	Complete data	Hidden variables / missing values
Known structure	**Count** Closed-form MLE or Bayesian estimate	**Expectation-Maximisation** Inference fills in the gaps, then count
Unknown structure	**Search over graphs** Independence tests, scores (BIC, BDeu), gradients	**Structural EM** Search and EM nested – the hardest case

A tiny triage network, ten patients



$P(I, F, C) = P(I) \cdot P(F \mid I) \cdot P(C \mid I)$

Parameters **θ**
All binary → 1 + 2 + 2 = **5 free parameters**

Patient	I	F	C
1	1	1	1
2	1	1	0
3	1	0	1
4	1	1	0
5	0	0	0
6	0	1	0
7	0	0	0
8	0	0	1
9	1	1	1
10	0	0	0

Learning parameters from complete data

The graph is known and every variable is observed. The answer will turn out to be: **count**.

Maximum Likelihood Estimation (MLE)

Likelihood: how well does θ explain D?

- **MLE** is a method of **estimating the parameters of a statistical model** given **observations**, by finding the parameter values that make the observed data **most probable** under the given model.
- **Maximize the likelihood** of making the observations given the parameters
- Consider the data $\mathbf{D = \{x_1, \ldots, x_n\}}$ that we believe comes from a model $\mathbf{p(x|\theta)}$ where **θ** are the model parameters
- We want to find the parameter **θ** that maximizes

$\mathbf{L(\theta; D)}$ - the *likelihood function* (of θ)

$$\boldsymbol{\theta = \text{argamx}_{\theta}\, p(D|\,\theta) = \text{argmax}_{\theta}\Pi_{i=1,n}p(x_i|\,\theta)}$$

Maximum likelihood estimation (MLE)

Likelihood: how well does θ explain D?

- We have multiple observations **generated i.i.d.** (independently and identical unknown distributions) D**=$\{x_1, x_2, \ldots, x_n\}$,** the likelihood function is

$$L(\theta; D) = L(\theta; x_1, x_2, \ldots, x_n) = \Pi_{i=1,n}\, p(x(i); \theta)$$

- It is easier to work with the log of the likelihood function - **log-likelihood**

$$L(\theta: D) = L(\theta; x_1, x_2, \ldots, x_n) = \Sigma_{i=1,n} \log\,(p(x(i); \theta))$$

- The likelihood function is invariant to permutation of data because we assumed the observations are drawn i.i.d (*independent and identically distributed*)
- This suggests that we should choose a representation of the data which does not depend on order
- One such representation is the **empirical distribution of data** which gives the relative frequency of different values of the random variables

MLE

If **p(x; θ)** is a *binomial distribution* with X=1 with probability **θ** and X=0 with probability **1- θ** , and unknown **θ** then

$$L(\theta;D) = \Sigma_k \, No_k \log \theta_k \quad \text{with } k=1,2$$

If **p(x; θ)** is a *multinomial distribution* over **M** values (X=a_1 with probability θ_1, X=a_2 with probability θ_2, … and $\Sigma_k \theta_k = 1$) and unknown **θ** (i.e., θ_1,.., θ_M) then

$$L(\theta;D) = \Sigma_k \, No_k \log \theta_k \quad \text{with } k=1,M$$

Find θ so as to maximize L(θ;D)

Kullback–Leibler divergence

MLE = getting as close as possible to the data

- We want to construct a probability distribution **p** as “close” as possible to the real one $\mathbf{p}^*$ of D
- How do we evaluate “closeness”?
- We will use the **Kullback–Leibler divergence**
- KL divergence is a measure of how one probability distribution diverges from a second probability distribution (relative entropy)

Kullback–Leibler divergence

$$KL(p^* \parallel p) = \sum_x p^*(x) \log \frac{p^*(x)}{p(x)} = \underbrace{-H(p^*) - \mathbb{E}_{x \sim p^*}\left[\log p(x)\right]}_{\text{Minimize}}$$

- H(p*) is the entropy of p*
- The first term does not depend on **p**
- Hence minimizing KL divergence is equivalent to <u>maximizing</u> the **expected log-likelihood**

$$\mathbf{E_{x \sim p^*}[\log p(x)]}$$

NB: Although we can now compare models, since we are not computing **H(p*)**, we don't know how close we are to the optimum

The same objective pre-trains every large language model: maximise log P(next token | context).

MLE example

Likelihood: how well does θ explain D?

5 tosses of a coin of which **3 are heads** ($x \in \{h,t\}$), following a distribution **p**

Assume that **$p(x=h) = \theta$** and **$p(x=t) = 1-\theta$**

If the observed data is **D = {h,h,t,h,t}** the likelihood is

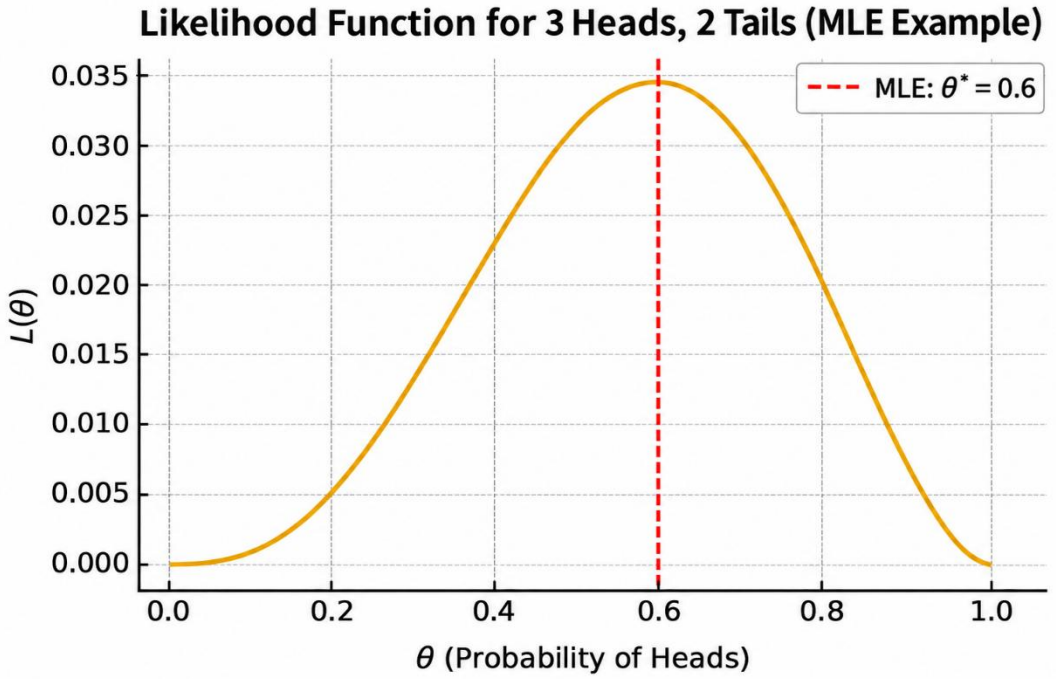
$$L(\theta) = \Pi_i p(x_i) = \theta \cdot \theta \cdot (1-\theta) \cdot \theta \cdot (1-\theta)$$

Maximize the log likelihood, $L(\theta) = 3\log\theta + 2\log(1-\theta)$

Find the maximum of this function by differentiating and setting the derivative to zero

$$3/\theta - 2/(1-\theta) = 0 \rightarrow \theta^* = 3/5$$

This gives $\theta^* = 0.6$

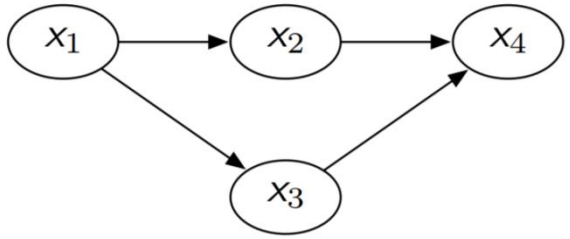


$L(\theta) = \theta^3(1-\theta)^2$, scaled to its maximum

MLE in Bayesian networks

- The previous discussion corresponds to a Bayesian network having one random variable, with a prior distribution.
- What if we have a BN with several random variables (nodes) and posterior distributions?
- We see now how parameter estimation for one-node graphical model extends for learning parameters of a general directed graphical model.

Example



$\mathbf{p(x_1,\ldots, x_4) = p(x_1)p(x_2|x_1)p(x_3|x_1)p(x_4|x_2, x_3)}$

The joint distribution can be represented with four sets of parameters $\boldsymbol{\theta_1, \ldots, \theta_4}$.

Each $\boldsymbol{\theta_i}$ defines a multinomial distribution corresponding to each joint assignment for the parents of $\mathbf{x_i}$

Suppose we have N samples of the 4-tuple $\mathbf{(x_1, x_2, x_3, x_4)}$.

The **likelihood function** can be written as

$$\mathbf{L(\theta;D) = \Pi_N\ p(x_1)p(x_2|x_1)p(x_3|x_1)p(x_4|x_2, x_3)}$$

The **log-likelihood function** can be written as

$$L(\theta;X) = \log p(x_1;\boldsymbol{\theta}_1) + \log p(x_2|x_1;\boldsymbol{\theta}_2) + \log p(x_3|x_1;\boldsymbol{\theta}_3) + \log p(x_4|x_2,x_3;\boldsymbol{\theta}_4)$$

- **Each individual term is the likelihood function of a multinomial distribution**
- We compute empirical conditional distribution for each variable $\mathbf{x_i}$ with respect to it's parents, e.g. **p(x2|x1)** for all possible values of $\mathbf{x_1}$.
- And this will be **the maximum likelihood estimation** of the Bayesian network

MLE in Bayesian networks

- Given a Bayesian network

$$\mathbf{p(x) = \Pi_{i=1,n}\theta_{xi|pa(xi)}}$$

$$P(X_i = x_i \mid Pa(X_i) = pa(x_i))$$

and i.i.d. samples $\mathbf{D=\{x^1, x^2, \ldots, x^m\}}$, $\mathrm{pa}(\mathrm{x}^{\mathrm{i}})$ – parents of x^{i}

- Compute the maximum likelihood estimate of the parameters (the CPD)
- The likelihood

$$L(\theta, D) = \prod_{i=1}^{n} \prod_{j=1}^{m} \theta_{x_i^{(j)} \mid pa(x_i^{(j)})}$$

$$\log L(\theta, D) = \log \left(\prod_{j=1}^{m} \prod_{i=1}^{n} \theta_{x_i^{(j)} \mid pa(x_i^{(j)})} \right)$$

Maximization of the (log) likelihood function decomposes into separate maximizations for the local conditional distributions

- Taking logs and combining like terms, this becomes

$$\log L(\theta, D) = \sum_{i=1}^{n} \sum_{x_i, pa(x_i)} \#(x_i, pa(x_i)) \cdot \log \theta_{x_i \mid pa(x_i)}$$

MLE in Bayesian networks

$$\log L(\theta, D) = \sum_{i=1}^{n} \sum_{x_i, pa(x_i)} \#(x_i, pa(x_i)) \cdot \log \theta_{x_i|pa(x_i)}$$

It can be shown that

$$\theta^*_{x_i \,|\, pa(x_i)} = \frac{\#(x_i,\, pa(x_i))}{\#(pa(x_i))}$$

Example

- Bayesian network
- Has 6 parameters



$$\theta_X^0 = P(X = 0)$$
$$\theta_X^1 = P(X = 1)$$
$$\theta_{Y|X}^0 = P(Y = 0 \mid X = 0)$$
$$\theta_{Y|X}^1 = P(Y = 1 \mid X = 0)$$
$$\theta_{Y|X'}^0 = P(Y = 0 \mid X = 1)$$
$$\theta_{Y|X'}^1 = P(Y = 1 \mid X = 1)$$

- Dataset

$$D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}$$

Likelihood

$$L(\theta; D) = \prod_{m=1}^{M} P(x_m \mid \theta) \cdot P(y_m \mid x_m, \theta)$$

$$= \prod_m \theta_X^{x_m} \prod_m \theta_{Y|X}^{y_m|x_m}$$

Example



- Expanded likelihood

$$L(\theta; D) = (\theta_X^0)^{M[0]} \cdot (\theta_X^1)^{M[1]} \cdot (\theta_{Y|X}^0)^{M[0,0]} \cdot (\theta_{Y|X}^1)^{M[0,1]} \cdot (\theta_{Y|X'}^0)^{M[1,0]} \cdot (\theta_{Y|X'}^1)^{M[1,1]}$$

$$\begin{aligned}
M[0] &= \text{number of samples with } X = 0, \\
M[1] &= \text{number of samples with } X = 1, \\
M[0,0] &= \text{number of samples with } X = 0, Y = 0, \\
M[0,1] &= \text{number of samples with } X = 0, Y = 1, \\
M[1,0] &= \text{number of samples with } X = 1, Y = 0, \\
M[1,1] &= \text{number of samples with } X = 1, Y = 1.
\end{aligned}$$

Example

- There are 3 sets of independent terms which can be separately maximized
- For X

$$\theta_X^{0*} = \frac{M[0]}{M[0] + M[1]}, \quad \theta_X^{1*} = 1 - \theta_X^{0*}$$



- For Y | X=0

$$\theta_{Y|X=0}^{0*} = \frac{M[0,0]}{M[0,0] + M[0,1]}, \quad \theta_{Y|X=0}^{1*} = 1 - \theta_{Y|X=0}^{0*}$$

- For Y | X=1

$$\theta_{Y|X=1}^{0*} = \frac{M[1,0]}{M[1,0] + M[1,1]}, \quad \theta_{Y|X=1}^{1*} = 1 - \theta_{Y|X=1}^{0*}$$

$$\theta^* = \frac{\text{count of configuration}}{\text{count of parent configuration}}$$

$$\begin{aligned}
M[0] &= \text{number of samples with } X = 0, \\
M[1] &= \text{number of samples with } X = 1, \\
M[0,0] &= \text{number of samples with } X = 0, Y = 0, \\
M[0,1] &= \text{number of samples with } X = 0, Y = 1, \\
M[1,0] &= \text{number of samples with } X = 1, Y = 0, \\
M[1,1] &= \text{number of samples with } X = 1, Y = 1.
\end{aligned}$$

Use of MLE

- Learning based on **MLE is used in many ML models** not only in Bayesian Networks

Probabilistic Graphical Models

Bayesian Networks

- Factorized as $P(X) = \prod_i P(X_i \mid \mathrm{Pa}(X_i))$
- Learn parameters by maximizing $\sum_n \log P(x^{(n)}|\theta)$

Markov Random Fields / Conditional Random Fields (CRFs)

- Undirected models
- Optimizes parameters of potential functions

Hidden Markov Models (HMMs)

  - For sequence data
- Uses the Expectation-Maximization (EM) algorithm to estimate transition and emission probabilities

Use of MLE

Latent Variable Models

Gaussian Mixture Models (GMMs)

- MLE maximizes $\sum_n \log \sum_k \pi_k \mathcal{N}(x^{(n)} | \mu_k, \Sigma_k)$
- Solved by EM algorithm

Factor Analysis / PCA (Probabilistic PCA)

- Probabilistic PCA can be derived as MLE on a Gaussian latent variable model

Latent Dirichlet Allocation (LDA)

- Topic model trained via MLE or Variational EM

Use of MLE

Generative Models in Deep Learning

Variational Autoencoders (VAEs)

- Train by maximizing an *approximate* likelihood (ELBO ≈ expected log-likelihood – KL term)

$$\text{ELBO} = \mathbb{E}_{q(z|x)}[\log p_\theta(x|z)] - KL(q(z|x)||p(z))$$

Diffusion Models

- Also explicit likelihood models
- Optimize a *variational lower bound* on log-likelihood during training

Autoregressive Models

- e.g. PixelCNN, WaveNet, language models
- Factorize $p(x) = \prod_i p(x_i|x_{<i})$
- Training = maximize log-likelihood → equivalent to minimizing cross-entropy loss

Use of MLE

Reinforcement learning

REINFORCE

- Some policy gradient methods (e.g. REINFORCE) are based on likelihood maximization

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta(\tau)}[\nabla_\theta \log p_\theta(\tau) R(\tau)]$$

Bayesian parameter learning

Treat θ as uncertain, and let the prior act as **imaginary data** seen before the real data.

Bayesian parameter learning - Intuition

The learning approaches we have discussed so far are *based on the principle of maximum likelihood estimation*.

While being extremely general, there are limitations of this approach

Example

- Modeling the outcome of a biased coin, **X={heads,tails}**
- We toss the coin **10 times**, observing **6 heads**.
- If **θ** denotes the **probability of observing heads**, the maximum likelihood estimate (**MLE)** is given by

$$\theta_{MLE} = \frac{num_heads}{num_heads + num_tails} = 0.6$$

- We continue tossing the coin such that after a **100 total trials** (including the 10 initial trials), we observe **60 heads**
- Again, we can compute the MLE as

$$\theta_{MLE} = \frac{num_heads}{num_heads + num_tails} = 0.6$$

- The maximum likelihood estimate does not change as we observe more data
- This seems counterintuitive - our ***confidence*** in predicting heads with probability **0.6** should be higher in the second setting where we have seen many more trials of the coin

Bayesian parameter learning - Intuition

- The reason why MLE fails to distinguish the two settings is due to an implicit assumption
- MLE assumes that the only source of uncertainty is due to the variables **X** and the quantification of this uncertainty is based on a fixed parameter $\boldsymbol{\theta}_{\mathbf{MLE}}$
- In contrast to maximum likelihood estimation, Bayesian learning explicitly models uncertainty over both the variables **X** and the parameters **θ**
- In other words, the model parameters **θ** are random variables as well
- A *prior* distribution over the parameters **p(θ)** encodes our initial beliefs
- These beliefs are subjective
- For example, we can choose the prior over **θ** for a biased coin to be uniform between 0 and 1
- If however we expect the coin to be fair, the prior distribution can be peaked around **θ=0.5**.

Bayesian parameter learning - Intuition

- Example
  - We are 90% sure that **p** is between 0.4 and 0.6
  - We are 95% sure that **p** is between 0.3 and 0.8
- We can model our prior belief using a *distribution of p*, as if **p** was random.
- In reality the true parameter is not random. However the Bayesian approach is a way of modeling our belief about the parameter as if it was random

- Therefore: Instead of choosing a single set of parameters $\boldsymbol{\theta}$, we consider a **probability distribution over the parameters**.

Basic Idea

- Instead of a fixed value $\boldsymbol{\theta}$, we treat the parameters $\boldsymbol{\theta}$ themselves as **random variables** with a **prior distribution p($\boldsymbol{\theta}$)**
- After observing the data D, we **update our knowledge**

  P($\boldsymbol{\theta}$ **|*D*)** about $\boldsymbol{\theta}$ using **Bayes' theorem**

$$P(\theta \mid D) = \frac{P(D \mid \theta)P(\theta)}{P(D)}$$

Bayesian parameter learning

- Observing data **D** in the form of evidence allows us to update our beliefs using Bayes' rule

$$p(\theta \mid D) = \frac{p(D \mid \theta)\, p(\theta)}{p(D)} \propto p(D \mid \theta)\, p(\theta)$$

$$posterior \propto likelihood \times prior$$

- Bayesian learning provides a principled mechanism for incorporating prior knowledge into the model
- This prior knowledge is useful in many situations such as when we want to provide uncertainty estimates about the model parameters or when the data available for learning a model is limited

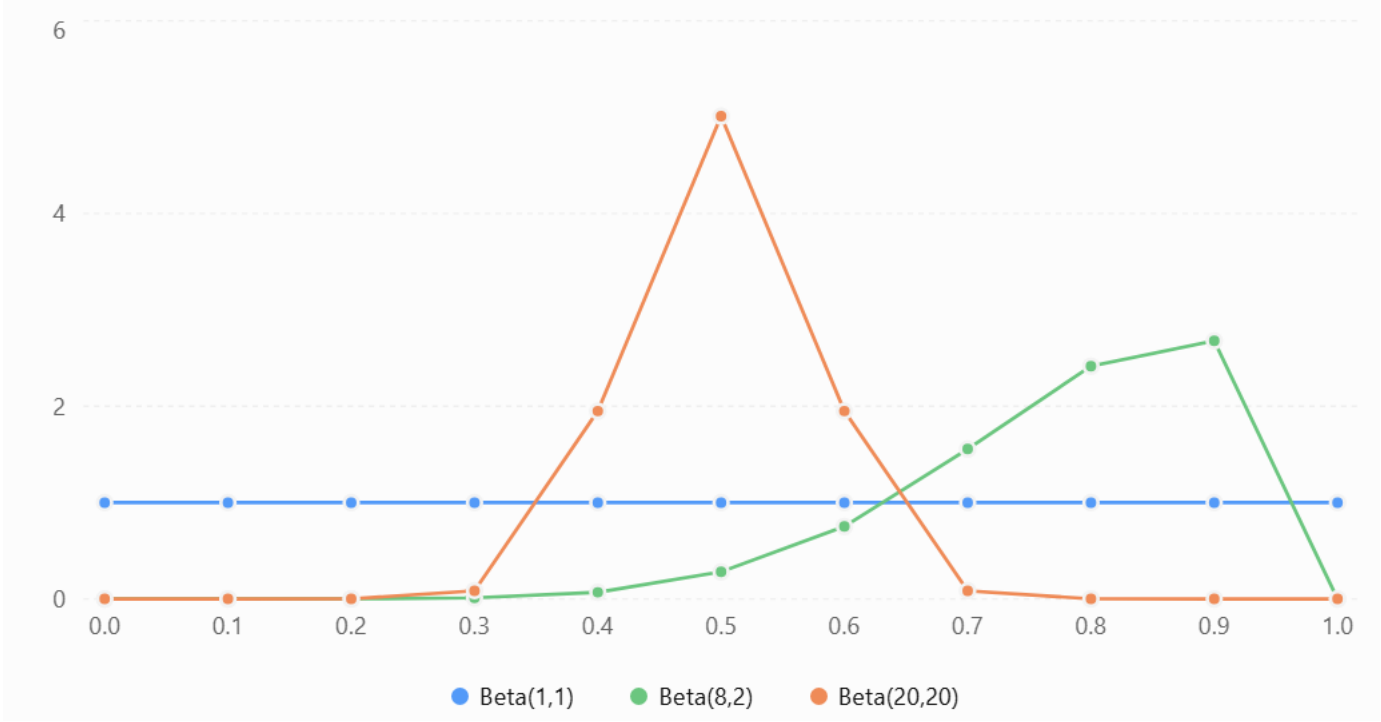
Prior distribitions

Before observing any data, what do we believe about the probabilities we are trying to learn?

Beta distribution — when there are two possible outcomes

$$\theta \sim \text{Beta}(\alpha, \beta)$$

- α: prior evidence in favor of heads.
- β: prior evidence in favor of tails.



Prior distribitions

Before observing any data, what do we believe about the probabilities we are trying to learn?

Dirichlet distribution — when there are more than two possible outcomes

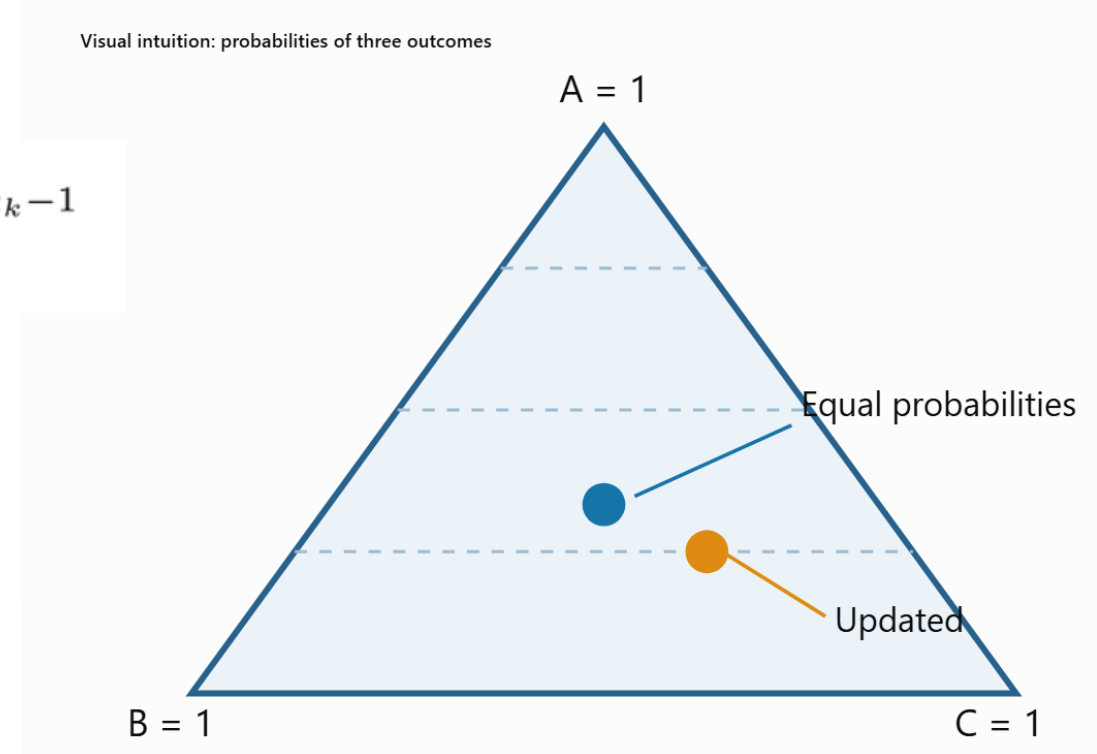
$$(\theta_A, \theta_B, \theta_C) \sim \text{Dirichlet}(\alpha_A, \alpha_B, \alpha_C)$$

$$P(\theta) = \text{Dirichlet}(\theta \mid \alpha) = \frac{1}{B(\alpha)} \prod_{k=1}^{K} P(X_j = k)^{\alpha_k - 1}$$

$$\mathrm{B}(\boldsymbol{\alpha}) = \frac{\prod_{i=1}^{K} \Gamma(\alpha_i)}{\Gamma\left(\sum_{i=1}^{K} \alpha_i\right)}, \qquad \boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K).$$

$\Gamma(n) = (n-1)!$ for positive integers

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx$$



Example with binary variables and Beta

- Let's say we have a coin and we want to learn the probability $\theta = P(Heads)$
- We choose a **prior** $P(\theta) = Beta(\alpha, \beta)$.
  • If $\alpha = \beta = 1$, we have a uniform distribution (no prior preference)
- We observe data:
  • 3 heads, 2 tails $\rightarrow h = 3, t = 2$
- The posterior becomes:

$$P(\theta \mid D) = Beta(\alpha + h, \beta + t)$$
$$P(\theta \mid D) = Beta(4,3)$$

which has a mean:

- $E[\theta] = \frac{4}{4+3} = 0.57$

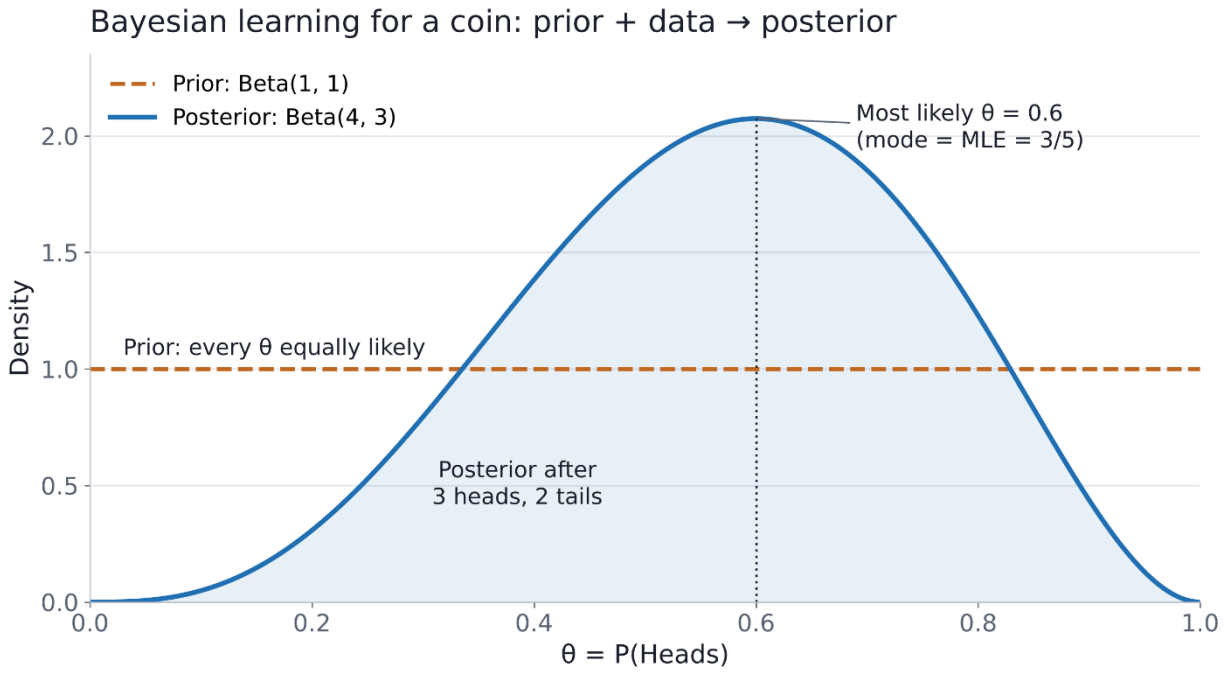
- Instead of a fixed estimate (0.6 as in MLE), we now have a **complete probability distribution** that reflects our **uncertainty** about θ

Example with binary variables and Beta

- The probability $\theta = P(Heads)$

$$P(\theta \mid D) = Beta(4,3)$$

- The **orange dashed line** is the prior $P(\theta) = Beta(1,1)$, which is uniform, meaning we initially have no preference
- The **blue curve** is the posterior $P(\theta \mid D) = Beta(4,3)$, which peaks around $\theta \approx 0.57$, showing our updated belief after observing the data 3 heads, 2 tails



Conjugate priors

- If the posterior distributions $p(\theta \mid x)$ are in the same probability distribution family as the prior probability distribution $p(\theta)$, the prior and posterior are then called **conjugate distributions,** and the prior is called a **conjugate prior for the likelihood function**
- When calculating posterior distribution using Bayes' rule, to calculate the denominator **p(D)**, we use the **conjugate prior**
- This is convenient because if we know the normalizing constant p(D), then we get the denominator in Bayes' rule

$$p(\theta \mid D) = \frac{p(D \mid \theta)\, p(\theta)}{p(D)} \propto p(D \mid \theta)\, p(\theta)$$

$$posterior \propto likelihood \times prior$$

Conjugate priors

The Beta distribution is **the conjugate prior** of the Bernoulli and binomial distributions

The Dirichlet distribution is **the conjugate prior** of the categorial and multinomial distributions

See Annex 1 for explanations about the Beta distribution

See Annex 2 for explanations about the Dirichlet distribution

Bayesian parameter learning

For **binary data**

The **prior** is a Beta distribution

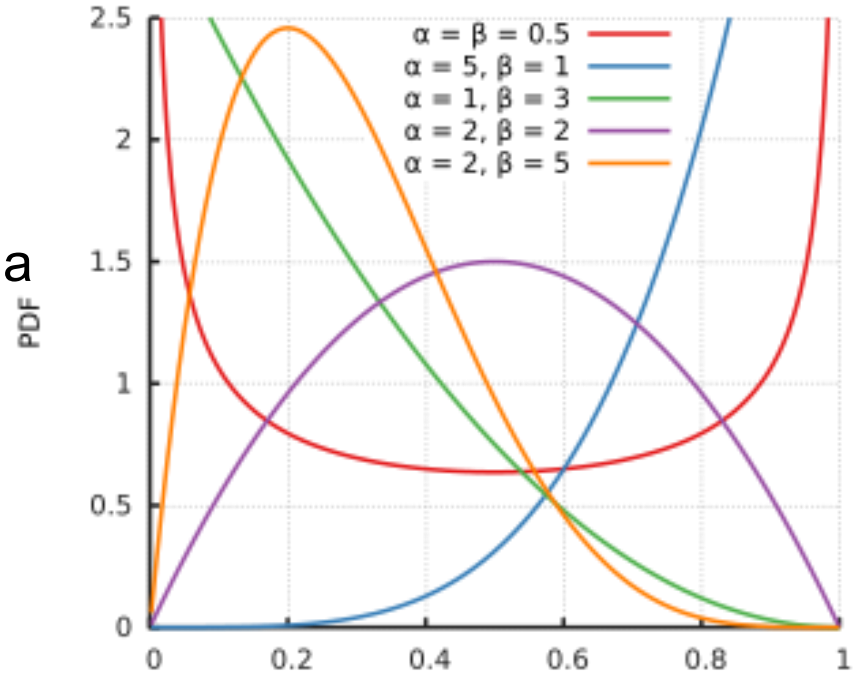
$$P(\theta) = \text{Beta}(\alpha, \beta)$$

The **posterior** is also a Beta distribution (because Beta is a conjugate prior of Bernoulli and binomial distributions)

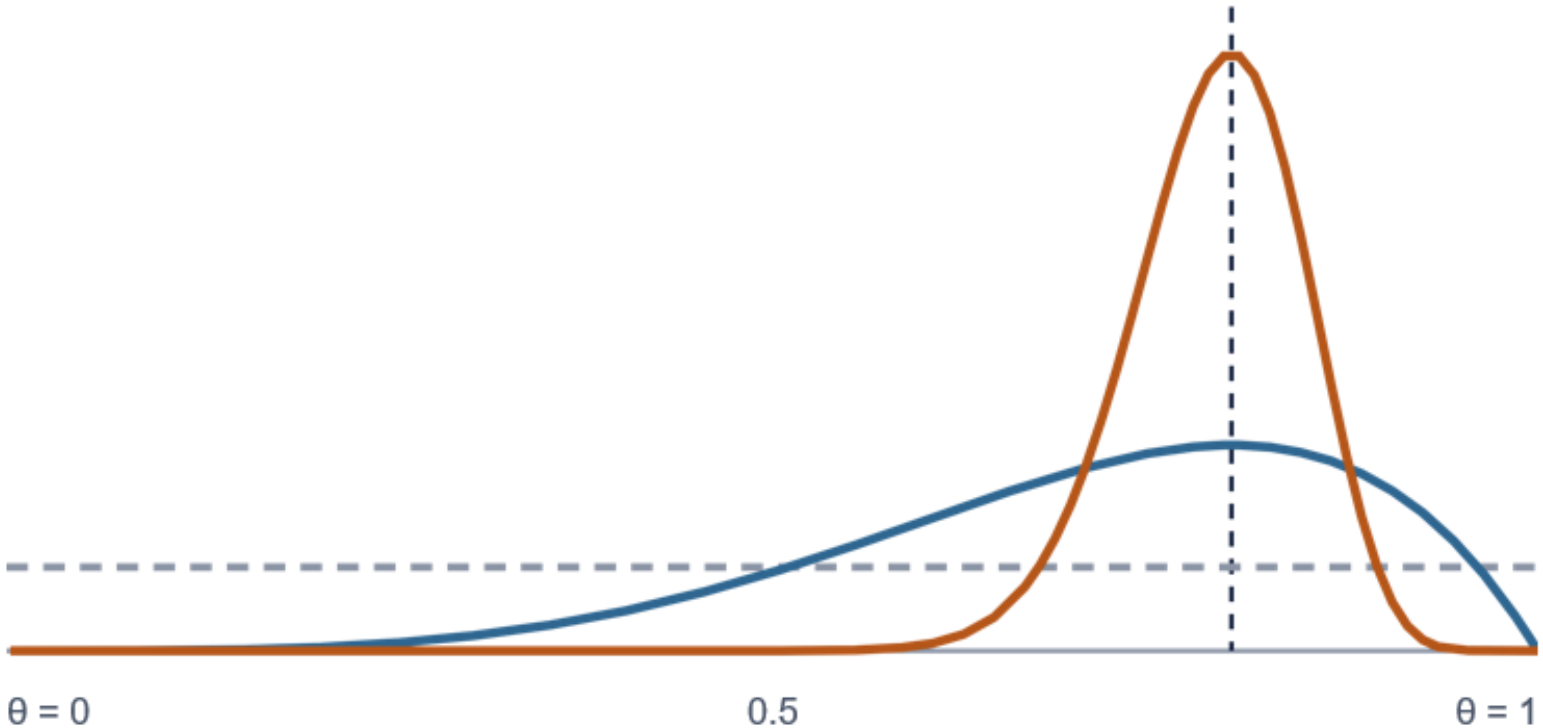
$$P(\theta \mid D) = \text{Beta}(\alpha + h,\ \beta + t)$$

h = the number o successes (heads)

t = number of failures (tail)



More data, sharper belief about θ



θ = P(F=1 | I=1)

Prior Beta(1, 1)

No opinion: all θ equally likely

4 of 5 with fever

Beta(5, 2) · mean 0.71 · wide

40 of 50 with fever

Beta(41, 11) · mean 0.79 · sharp

MLE = 0.8 both times

It cannot tell the cases apart

Bayesian parameter learning

- We can now extend the binary model to its categorical generalization
- Instead of being limited to binary outcomes, we can now consider the categorical dataset of a **K-sided dice** rolled **N** times.
- Let $\mathbf{D=\{X_1=k_1,...,X_N=K_N\}}$ where $X_j \in \{1,...,K\}$ for the jth outcome.
- The parameterization of the model is $\mathbf{\theta=(P(X_j=1),...,P(X_j=K))}$, which denotes the probability of each outcome, and where $\sum_{k=1,K}\mathbf{P(X_j=k)=1}$.

Bayesian parameter learning

- The likelihood of observing our dataset given a specific parameterization is

$$P(\mathcal{D} \mid \theta) = \prod_{k=1}^{K} P(X_j = k)^{\sum_{j=1}^{N} \mathbf{1}\{X_j = k\}}$$

$$\mathbf{1}\{X_j = k\} = \begin{cases} 1, & \text{if } X_j = k \\ 0, & \text{if } X_j \neq k \end{cases}$$

an indicator function that equals 1 when X_j=k and 0 otherwise

$$N_k = \sum_{j=1}^{N} \mathbf{1}\{X_j = k\}$$

N_k is the number of times outcome k appears in the dataset of N observations.

Multinomial likelihood

- Each observation contributes to the likelihood of its corresponding category
- The more often an outcome occurs, the higher the exponent of its probability

Bayesian parameter learning

For **categorical data**

The **prior** is a Dirichlet distribution

$$P(\theta) = \text{Dirichlet}(\alpha_1, \alpha_2, ..., \alpha_K)$$

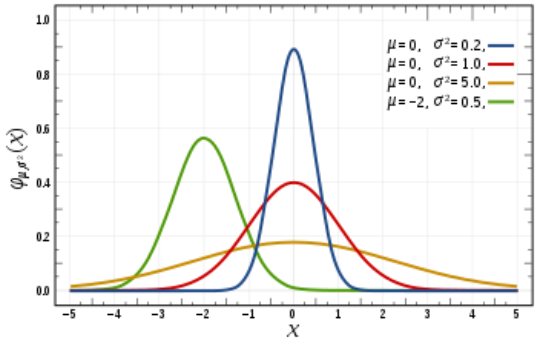
The **posterior** is also a Dirichlet distribution (because Dirichlet is a conjugate prior of categorial and multinomial distributions)

$$P(\theta \mid D) = \text{Dirichlet}(\alpha_1 + N_1, \alpha_2 + N_2, ..., \alpha_K + N_K)$$

Annex 1: Beta distribution

$$P(\theta \mid D) = \frac{P(D \mid \theta)P(\theta)}{P(D)}$$

- We need a distribution for **P(θ)** in [0,1]
- We can not choose a Gaussian distribution as it is not in [0,1]

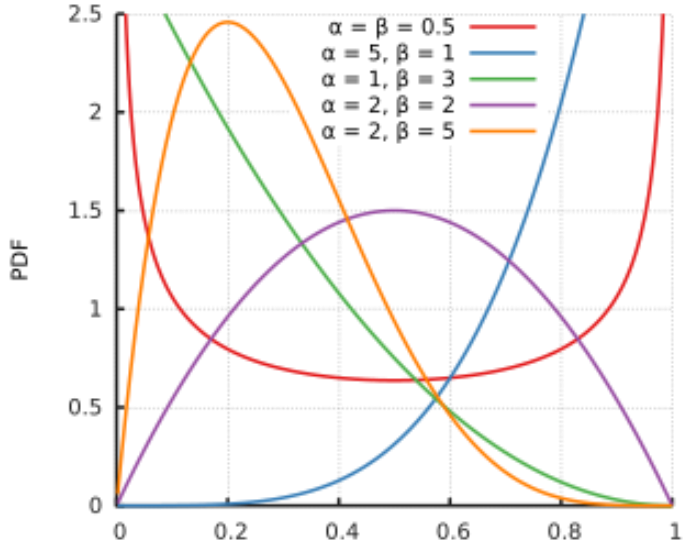


- Everybody is using the **Beta distribution**

Beta distribution

- A family of continous probability distributions in [0,1] parameterised by **α, β >0** (shape parameters that control the shape of the distribution). The **pdf** is

$$
\begin{aligned}
f(x;\alpha,\beta) &= \text{constant} \cdot x^{\alpha-1}(1-x)^{\beta-1} \\
&= \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\int_0^1 u^{\alpha-1}(1-u)^{\beta-1}\,du} \\
&= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\,x^{\alpha-1}(1-x)^{\beta-1} \\
&= \frac{1}{\mathrm{B}(\alpha,\beta)}x^{\alpha-1}(1-x)^{\beta-1}
\end{aligned}
$$



$\Gamma(n) = (n-1)!$ for positive integers

$$\Gamma(z) = \int_0^\infty x^{z-1}e^{-x}\,dx$$

Beta distribution

- Beta distribution (function) can be extended to functions with more than 2 arguments – multivariate Beta function

$$\mathrm{B}(\alpha_1, \alpha_2, \ldots \alpha_n) = \frac{\Gamma(\alpha_1)\,\Gamma(\alpha_2)\cdots\Gamma(\alpha_n)}{\Gamma(\alpha_1 + \alpha_2 + \cdots + \alpha_n)}$$

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\, dx$$

Example

- **Example**
- Suppose we are given a sequence of **N** coin tosses, **$D=\{X_1,...,X_N\}$**
- We want to infer the probability of getting heads which we denote by **θ**.
- We can model this as a sequence of Bernoulli trials with parameter θ. A natural conjugate prior in this case is the beta distribution with

$$P(\theta) = Beta(\theta \mid \alpha_H, \alpha_T) = \frac{\theta^{\alpha_H - 1}(1-\theta)^{\alpha_T - 1}}{B(\alpha_H, \alpha_T)}$$

where the normalization constant **B(·)** is the beta function

Example

$$P(\theta) = Beta(\theta \mid \alpha_H, \alpha_T) = \frac{\theta^{\alpha_H - 1}(1-\theta)^{\alpha_T - 1}}{B(\alpha_H, \alpha_T)}$$

- **$\alpha=(\alpha_H,\alpha_T)$** are called the hyperparameters of the prior
- The expected value of **θ** is **α_H / $(\alpha_H+\alpha_T)$**
- The sum of the hyperparameters **$(\alpha_H+\alpha_T)$** can be interpreted as a measure of confidence in the expectations they lead to.
- Intuitively, we can think of **α_H** as the number of heads we have observed before the current dataset.

Example

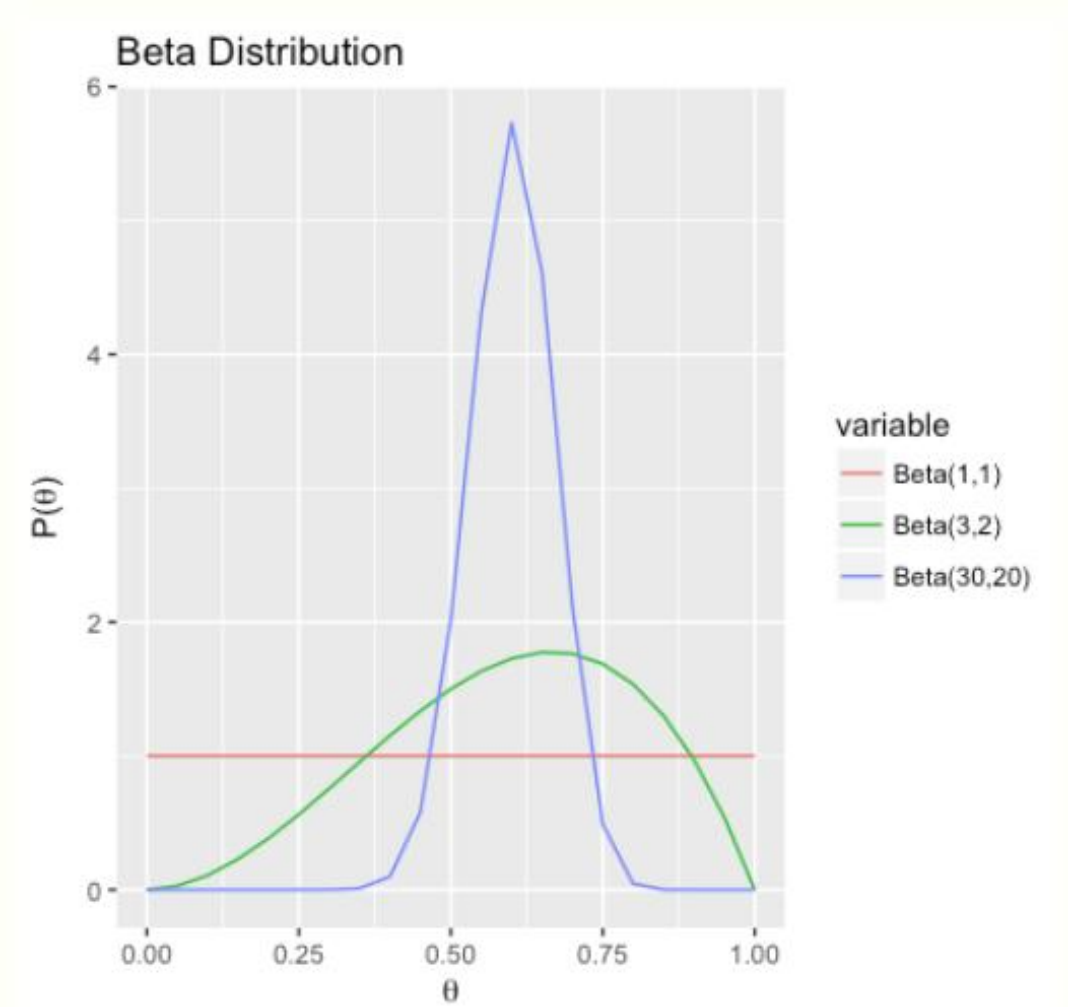
- Out of **N** coin tosses, if the number of heads and the number of tails are $\mathbf{N_H}$ and $\mathbf{N_T}$ respectively, then it can be shown that the posterior is:

$$P(\theta \mid N_H, N_T) = \frac{\theta^{N_H+\alpha_H-1}(1-\theta)^{N_T+\alpha_T-1}}{B(N_H+\alpha_H,\, N_T+\alpha_T)}$$

which is a Beta distribution with parameters $\mathbf{(N_H+\alpha_H)(N_T+\alpha_T)}$**.**

- We can use this posterior distribution as the prior for more samples with the hyperparameters by adding each extra piece of information as it comes from additional coin tosses.

Example



The exponents (3,2) and (30,20)
can both be used to encode
the belief that θ =0.6

But the second set of exponents
imply a stronger belief
as they are based on a larger sample

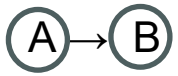
Annex 2: A complete example

Example: Learning using conjugate priors to update beliefs based on observed data

- A Bayesian network with two binary variables:

  A - represents whether it is raining (A=1) or not raining (A=0)

  B - represents whether the grass is wet (B=1) or dry (B=0)



- We want to learn the conditional probability **P(B=1|A)** using a conjugate prior for the probability of wet grass given rain or no rain.

Step1 - Initial

- **Prior Information**: We start with a ***Beta prior*** for each conditional probability:
  - For P(B=1|A=1), we use Beta(α_1,β_1) with parameters α_1=2 and β_1=2
  - For P(B=1|A=0), we use Beta(α_0,β_0)) with parameters α_0=1 and β_0=1
- **We observe the data:** whether it rained and whether the grass was wet
  - (A=1,B=1),(A=1,B=0),(A=1,B=1),(A=0,B=0),(A=0,B=1),(A=1,B=1),(A=0,B=0)
- **We have to** update the prior distributions based on this data to find the posterior distributions for **P(B=1|A=1)** and **P(B=1|A=0)**

Why the specific values for the parameters α and β?

- Remember that Beta(α,β) is a common conjugate prior for binary (Bernoulli) data. In this case it represents the probability of success P(B=1)
- Parameters α and β represent "pseudo-counts" or "prior evidence"
  - α represents prior evidence for successes (the prior belief about the number of times B=1 has been observed)
  - β represents prior evidence for failures (the prior belief about the number of times B=0 has been observed)
  - **For P(B=1|A=1)**
    - The parameters $\alpha_1=2$ and $\beta_1=2$ represent a prior belief that we have observed two "successes" and two "failures" when A=1
    - These values reflects a weakly informative prior: provides some prior information (we believe there's a 50-50 chance of B=1 when A=1), but it isn't overly strong - the observed data has an impact on the posterior
  - **For P(B=1|A=0)**
    - The parameters $\alpha_0=1$ and $\beta_0=1$ represent an even weaker prior belief (one success and one failure). This choice gives less weight to prior beliefs, allowing the observed data for A=0 to have even more influence on the posterior
- Therefore, we start with an initial assumption and let the observed data predominantly shape the posterior

Observed data

(A=1,B=1)
(A=1,B=0)
(A=1,B=1)
(A=0,B=0)
(A=0,B=1)
(A=1,B=1)
(A=0,B=0)

Step 2 - Separate Observations by Condition

- When A=1 observations are (B=1),(B=0),(B=1),(B=1)
  - Count of B=1 given A=1: $n_{11}=3$
  - Count of B=0 given A=1: $n_{10}=1$
- When A=0 observations are (B=0),(B=1),(B=0)
  - Count of B=1 given A=0: $n_{01}=1$
  - Count of B=0 given A=0: $n_{00}=2$

Step 3 - Calculate the Posterior Parameters

- For P(B=1|A=1):
  - Update α_1 and β_1 based on observed counts: $\alpha_{1,posterior} = \alpha_1+n_{11} = 2+3 = 5$

    $\beta_{1,posterior} = \beta_1+n_{10} = 2+1 = 3$
  - The posterior for P(B=1|A=1) is *Beta(5,3)*
- For P(B=1|A=0)
  - Update α_0 and β_0 based on observed counts: $\alpha_{0,posterior} = \alpha_0+n_{01} = 1+1 =2$

    $\beta_{0,posterior} = \beta_0+n_{00} = 1+2 = 3$
  - The posterior for P(B=1|A=0) is *Beta(2,3)*

Step 4 – Update beliefs

- After observing the data, we have updated beliefs:
  - **P(B=1|A=1)** ~ **Beta(5,3)** with an expected value of

    E[P(B=1|A=1)] = 5/(5+3) = 5/8 = 0.625
  - **P(B=1|A=0)** ~ **Beta(2,3)** with an expected value of

    E[P(B=1|A=0)] = 2/(2+3) = 2/5 = 0.4

Therefore

P(B=1|A=1) ≈ 0.625

P(B=1|A=0) ≈ 0.4

Annex 3 : Dirichlet distribution

- In order to compute the prior for catgorial and multinomial, we need a **Dirichlet distribution**
- **Dirichlet(θ |α)** is a family of continous probability distributions parameterized by a vector **α** of positive reals. It is a multivariate generalisation of the Beta distribution
- The

$$f\left(x_1, \ldots, x_K; \alpha_1, \ldots, \alpha_K\right) = \frac{1}{\mathrm{B}(\boldsymbol{\alpha})} \prod_{i=1}^{K} x_i^{\alpha_i - 1}$$

$$\sum_{i=1}^{K} x_i = 1 \text{ and } x_i \geq 0 \text{ for all } i \in [1, K]$$

$$\mathrm{B}(\boldsymbol{\alpha}) = \frac{\prod_{i=1}^{K} \Gamma(\alpha_i)}{\Gamma\left(\sum_{i=1}^{K} \alpha_i\right)}, \qquad \boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K).$$

Bayesian parameter learning with Dirichlet

- **The Dirichlet distribution is the conjugate prior of the categorial distribution and multinomial distributions**
- We consider a categorical model and the Dirichlet distribution with hyperparameters $\boldsymbol{\alpha=(\alpha_1,...,\alpha_K)}$, indicating the number of observations of each outcome.
- Letting **α** be the "virtual" counts of the K outcomes before we observe the dataset, the prior is

$$P(\theta) = \text{Dirichlet}(\theta \mid \alpha) = \frac{1}{B(\alpha)} \prod_{k=1}^{K} P(X_j = k)^{\alpha_k - 1}$$

where **B(·)** is a normalization factor

Bayesian parameter learning with Dirischlet

- Because we use a Dirichlet prior, the posterior is also a Dirichlet distribution, and is:

$$P(\theta \mid \mathcal{D}) \propto P(\mathcal{D} \mid \theta)\, P(\theta) \propto \prod_{k=1}^{K} P(X_j = k)^{\sum_{j=1}^{N} \mathbf{1}\{X_j = k\} + \alpha_k - 1}$$

- This is equivalent to a Dirichlet distribution with updated counts **α′**

$$P(\theta \mid \mathcal{D}) \propto \text{Dirichlet}(\theta \mid \alpha')$$

where the updated count α′ is given by

$$\alpha'_k = \underbrace{\sum_{j=1}^{N} \mathbf{1}\{X_j = k\}}_{\text{observed data count}} + \underbrace{\alpha_k}_{\text{prior virtual count}}$$